# Chunghwan Lee

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**Research Interests**

- Reconstructing 3D scenes
- Privacy-preserving visual localization
- Neural fields in 3D visual computing

**Education**

**Hanyang University (HYU)** — Seoul, Korea
*M.S., Dept. of Electronic Engineering* (*GPA: 4.33/4.5 (until 2$^{nd}$sem.)*) — *Mar. 2022 – Present*
Advisor: Prof. Je Hyeong Hong

**Hanyang University (HYU)** — Seoul, Korea
*B.S., Dept. of Electronic Engineering* (*GPA: 3.82/4.5*) — *Mar. 2015 – Feb. 2022*
Thesis: *Deepfake Detection Using Supervised Temporal Feature Extraction Model and LSTM*
Advisor: Prof. Kijung Yoon

**Publications (International)**

1.**"Paired-Point Lifting for Enhanced Privacy-Preserving Visual Localization"**, **Chunghwan Lee**, Jaihoon Kim, Chanhyuk Yun, Je Hyeong Hong, *The IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023*

**Publications (Domestic)**

1."Deepfake Detection Using Supervised Temporal Feature Extraction Model and LSTM", **Chunghwan Lee(*)**, Jaihoon Kim(*), Kijung Yoon, *The Korean Institute of Broadcast and Media Engineers Fall Conference, 2022*

2."Weather Classification and Image Restoration Algorithm Attentive to Weather Conditions in Autonomous Vehicles", Jaihoon Kim(*), **Chunghwan Lee(*)**, Sangmin Kim, Jechang Jeong, *The Korean Institute of Broadcast and Media Engineers Fall Conference, 2021*
(** equal contribution)*

**Patents**

(*Korea*) "Localization method and apparatus using line cloud map data, line cloud map data generation method", Je Hyeong Hong, **Chunghwan Lee**, Chanhyuk Yun, *Registration No. 10-2023-0008666 (applicant: HYU)* — *Jan. 2023*

**Research Projects**

**HY-Vision Lab, Hanyang University (HYU)** — Seoul, Korea
*Enhancing Privacy-Preserving Visual Localization* — *Feb. 2022 – Present*

- Proposed a privacy preserving 3D line cloud generation method which draw lines by pairing two points, named paired-point lifting(PPL).
- Our method preclude the scene recovery attack that intrudes privacy without compromising camera pose re-localization accuracy.

*Deep Learning-based Sketched Face Recognition* — *Oct. 2021 – Feb.2022*

- Developed human sketched face recognition model by fine-tuning the ARKFACE and conducting an augmentation on sketch data with CycleGAN.

**NIAI Lab, Hanyang University (HYU)** — Seoul, Korea
*Deepfake Video Detection* — *Feb. 2021 – Sep. 2021*

- Developed the classification model discerning deepfake videos by training the model to extract the abrupt features.

**ICSP Lab, Hanyang University (HYU)** — Seoul, Korea
*Image-based Weather Classification and Restoration* — *Jun. 2020 – Oct. 2020*

- Designed a one-for-all model that processes from classification to reconstruction from bad weather artifacts; Derain, Desnow, Dehaze.

**Skills** **Programming** proficient in: Python (including libraries such as Numpy, Pandas, OpenCV, Open3D), PyTorch, Keras and Tensorflow (including visualization and debugging tools such as Tensorboard and Weights & biases); familiar with: C/C++, Java, MATLAB

**Honors**

**Third Place**, IC-PBL+ Graduate Championship 2022-2, IC-PBL Teaching and Learning Center, HYU, Seoul, Korea *2023*

**Excellence Award**, Undergraduate Hanyang Academic, HYU, Seoul, Korea *2022*

**Excellence Award**, The Korean Institute of Broadcast and Media Engineers Fall Conference 2021, The KIBME, Seoul, Korea *2021*

**Scholarships**

**HY-in Scholarships**, Graduate Hanyang University, Seoul, Korea *2022 – present*
$ 12,200

**Dept. E.E. Scholarship**, Undergraduate Hanyang University, Seoul, Korea *2015 – 2016*
$ 14,500

**Additional** **Paper reviewer - ICIP 2022-2023, WACV 2023-2024**